

MoTrPAC human-precovid-sed-adu data release notes – analysis

Data collection type: analysis
Levels included: Levels 2–4 (analysis results)
Next anticipated release: 2026-06-01, to include instructions for individual level data access

1. Overview & Quick Reference

This document describes the **human-precovid-sed-adu analysis** data collection and focuses on **Levels 2–4** ([see detailed description of data levels](#)) analysis results derived from the corresponding human pre-suspension (before the Covid-19 pandemic) quantification and identification results. In brief, **Level 2 (qc-norm)** provides quality-controlled normalized data used as feature-by-sample matrices for downstream analysis; **Level 3 (da)** provides **differential abundance (DA)** results (models/contrasts with effect sizes and multiple-testing adjustments); and **Level 4** provides downstream interpretation resources (e.g., enrichment and integrative outputs) when present. Key details are provided below:

- **Data collection name / code:** MoTrPAC pre-suspension (pre-COVID) human sedentary adults / human-precovid-sed-adu (PreCAWG).
- **Species / cohort / age:** *Homo sapiens* / adult sedentary participants, pre-COVID / ~18–73 years.
- **Current data collection version:** v1.3
- **Recommended version for new analyses:** v1.3.
- **Last release update:** 2025-05-07 (known issues update in the consortium release readme).
- **Data Levels included:** Levels 2–4 (analysis results).
- **Restrictions:**
 - Open access: Level 3 (da) and 4 (integrative)
 - **Data access restriction: Level 2 (qc-norm) and any associated phenotypic data. Individual level data is not openly available at this time. Access can be obtained via data access committee approval. Please inquire at motrpac-data.org for access.**
- **Dataset structures:**
 - Level 2 qc-norm matrices (plus sample/feature metadata) organized by ome and tissue;
 - Level 3 differential abundance (DA) tables
 - Level 4 enrichment / integrative outputs (e.g., CAMERA and clustering-based enrichment results).
 - A centralized feature map (**HUMAN_FEATURE_TO_GENE**) supports feature-level joins across omics and downstream annotation.

- **Scope:**

Metric	Value
Tissues	3 overarching categories (adipose , blood , muscle) covering tissue codes t02 , t03 , t04 , t05 , t06 , t07 , t10 , t11
Assays / platforms	Multiple ome/platform identifiers are supported by the package loader functions (including epigenomics); 40 qc-norm *_QC objects are distributed as packaged R data objects in

Metric	Value
	MotrpacHumanPreSuspensionData covering transcriptomics, proteomics, and metabolomics (targeted + untargeted).
Participants	175 unique pid values represented across packaged qc-norm objects with at least one multi-omic measurement
Samples	13,778 unique vialLabel values represented across packaged qc-norm objects

- **Time points (acute bout):** **pre_exercise**, **during_20_min**, **during_40_min**, **post_10_min**, **post_15_30_45_min**, **post_3.5_4_hr**, **post_24_hr** (not all omics/tissues include all time points; see assay-specific metadata).

- **Key identifiers (see Glossary for details):**

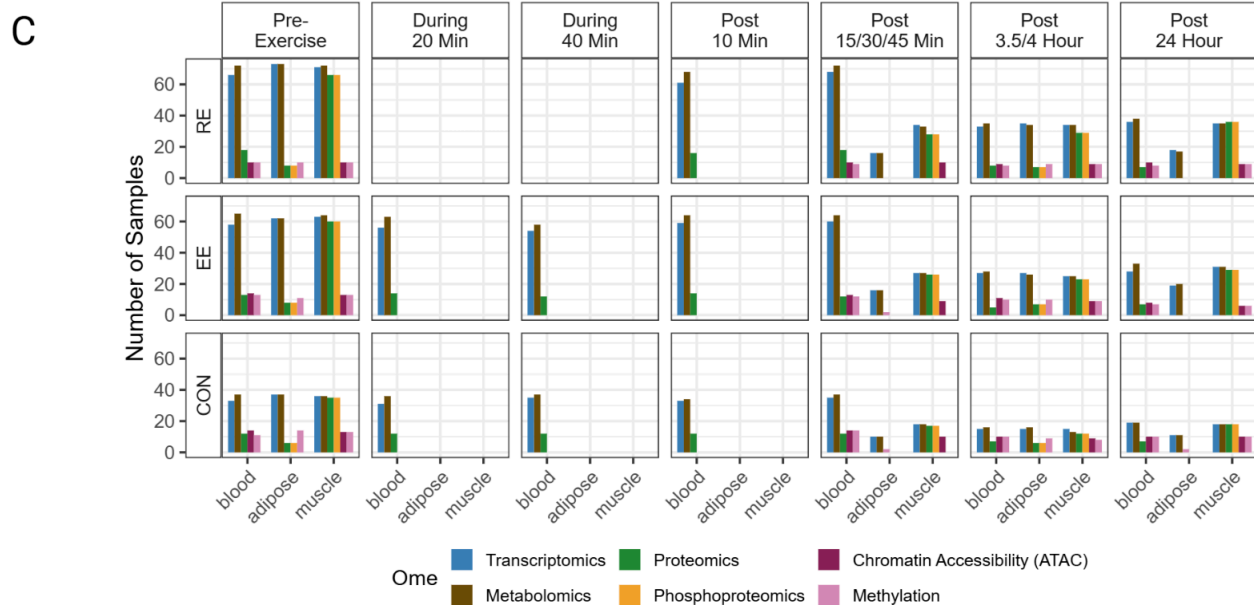
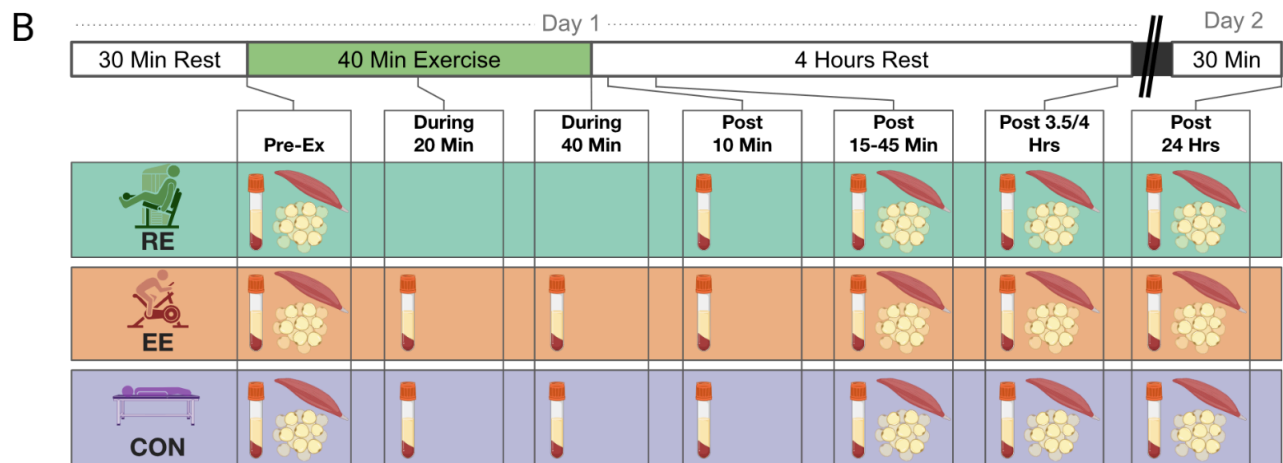
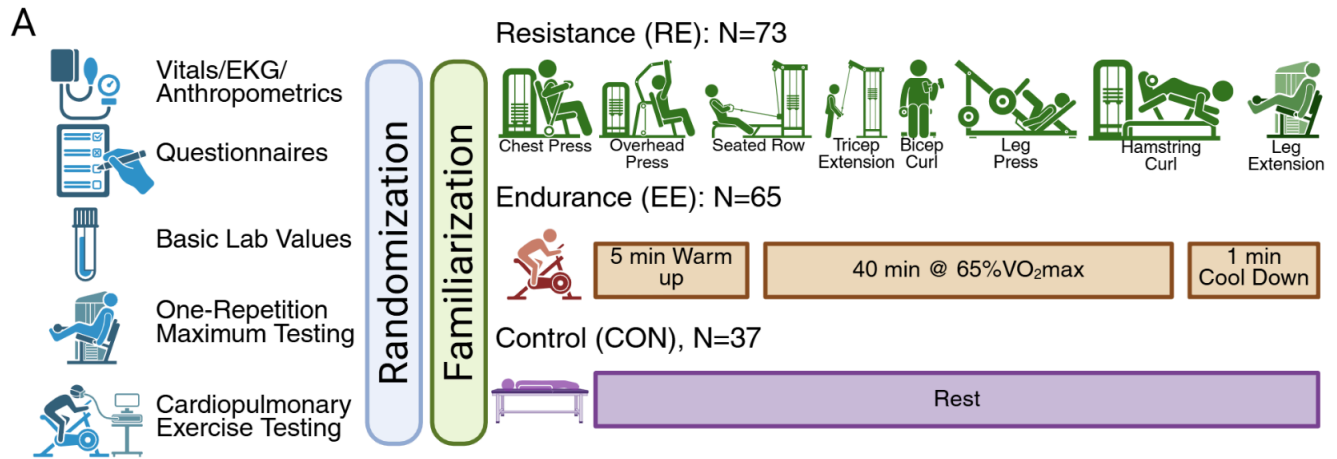
Identifier	Meaning
pid	Participant identifier (primary key for participant-level joins; used throughout clinical/phenotype tables).
vialLabel	Biospecimen (vial/sample) identifier used as sample IDs in Level 2 matrices (matrix columns) and as a join key between sample-level metadata and clinical/phenotype tables.
tissue_code	Tissue code token (e.g., t06-muscle , t02-plasma) used in GCS file names; note that some workflows also use coarse tissue categories (muscle , blood , adipose).
assay / ome	Assay/platform identifier (e.g., transcript-rna-seq , prot-pr , metab-u-rpneg) used in file names, object names, and filtering.
feature_id	Feature identifier used in Level 3 DA tables for feature-level joins and downstream annotation (see HUMAN_FEATURE_TO_GENE ; assay-dependent coverage).

- **Data locations:**
 - **Data Portal:** <https://motrpac-data.org> (after login, click “Pre-COVID Human Data Visualization” to access internal pre-COVID human resources).
 - **Zenodo DOI:** Not yet available
 - **R packages (recommended for convenient access and helper functions):**
 - **MotrpacHumanPreSuspensionAnalysis** (release pending).
- **Documentation and vignettes:**
 - Consortium release readme: [general_reference_docs/Consortium_Release_human-precovid-sed-adu-v1.2_2025FEB28.md](#)
 - Analysis collection description: [general_reference_docs/Data-Collections_Analysis-Results.md](#)

- Level definitions and conventions:
[general_reference_docs/Data-Collection_Release-Guidelines.md](#)
- Data-structure conventions (incl. PreCAWG naming):
[general_reference_docs/Guidelines-MoTrPAC Data Structure.md](#)

2. Narrative Summary

The human **pre-COVID pre-suspension** data include an adult sedentary cohort whose multi-omics data are permitted for analysis under DSMB constraints. The analysis collection focuses on the acute bout response by comparing timepoints relative to pre-exercise baseline and comparing changes in exercise groups (endurance [EE], resistance [RE]) to matched changes in controls (CON) (“difference in changes”). Below is a visual overview of the study and the available data. See publication for details.



(A) Overview of the acute exercise bout. Baseline testing is completed before randomization. After randomization, a familiarization period for exercise is completed. Patients arrive in the morning, fasted, to complete their respective interventions. Created in <https://BioRender.com>.

(B) Schematic of muscle, blood, and adipose biospecimen collection. No samples are collected during resistance exercise. Created in <https://BioRender.com>.

(C) Number of samples measured in each omic category (on any assay) by tissue, exercise modality and timepoint. See Supplementary Figure S1 to see overlapping measurements for all participants. ATAC = Assay for Transposase-Accessible Chromatin

The **analysis collection** transforms processed omics results into analysis-ready matrices and statistical results. As described in the quick reference, data objects are available for download from the Data Hub or can be accessed via R package (release pending) depending on restrictions. The data structure is the same for both access locations.

- **Level 2 (qc-norm)**: Quality-controlled, normalized feature-by-sample matrices (columns are `vialLabel`; feature identifiers are assay-dependent) with associated feature and sample metadata, suitable for custom exploratory analyses and modeling.
- **Level 3 (da)**: Differential abundance tables with standardized contrasts (within-group changes and between-group delta-delta comparisons) including effect sizes, confidence intervals, and multiple-testing adjusted p-values.
- **Level 4 (when present)**: Downstream interpretation and integration outputs, including enrichment results (e.g., CAMERA-based and clustering-based enrichment summaries) and reference resources (e.g., `HUMAN_FEATURE_TO_GENE`, metabolomics CV filtering resources). **These results are NOT available for download from the Data Hub, and must be accessed via the R package.**

Analysis essentials, see publication for details

Core Model: A linear mixed effects model was fit for each molecular feature using the dream function from the variancePartition R package. For count-based data (transcriptomics, ATAC-seq), voomWithDreamWeights was applied first.

Primary Comparison: The main results were generated using a "difference-in-changes" model. This compares the change from a pre-exercise baseline to a later timepoint within an exercise group (resistance or endurance) *against* the corresponding change in the non-exercising control group. These contrasts are specified in the DA tables.

Model Covariates: The final models included the following terms:

- Fixed Effects: group (RE, EE, CON), timepoint (as a cell-means interaction with group), clinical site, age, sex, and BMI.
- Random Effect: participant id was included to account for repeated measures from the same individual.
- Note: Race, ethnicity, and genetic ancestry were evaluated but not included as they did not improve model fit (based on BIC).

Model Selection:

The dream model was selected after a large-scale simulation study showed it provided the best balance of high statistical power while maintaining the target Type 1 error rate (FDR $\approx$ 5%) compared to six other common methods.

Additional Comparisons: Two other types of comparisons were made:

- Group-Specific: Change from baseline within a single exercise group (e.g., RE post-exercise vs. RE pre-exercise).
- RE vs. EE: A direct comparison of the change from baseline in the resistance group versus the endurance group.

Data Filtering: For a feature to be included in an analysis, a minimum of 3 participants with paired pre- and post-exercise samples were required in each group being compared. Features not meeting this criterion were excluded from differential analysis.

Significance Threshold: P-values for all comparisons were adjusted using the Benjamini-Hochberg method. A feature was considered statistically significant at a False Discovery Rate (FDR) of 0.05.

For users new to MoTrPAC analysis results

- Start with **Level 2 qc-norm** when you want a cleaned, analysis-ready matrix for exploratory analysis, visualization, and custom modeling.
- Use **Level 3 da** tables when you want standardized differential analysis results (effect sizes, p-values/FDR) for predefined contrasts/tests.
- Use the packaged phenotype/clinical metadata (e.g., **pheno** in R package) to subset and annotate samples (e.g., group assignment, sex, time point).

For users familiar with other MoTrPAC analysis releases

- The release is tied to QC data freeze v1.3; see Section 4 for full version history, known issues, and compatibility notes.
- This collection is a DSMB-constrained subset: only the adult sedentary cohort is permitted for downstream analysis; other concurrently generated groups may exist internally for QC but are not part of the official analysis release.

2.1 Experimental factors

- Cohort: adult sedentary participants (pre-suspension / pre-COVID phase).
- Randomized group: **ADUEndur** (EE, endurance), **ADUResist** (RE, resistance), **ADUControl** (CON, control).
- Timepoints (acute bout; see Section 1): **pre_exercise**, during-exercise draws, and multiple post-exercise timepoints up to **post_24_hr** (assay/tissue dependent). See timepoints above.
- Tissues: analysis uses three overarching categories (**blood**, **muscle**, **adipose**) mapped from MoTrPAC tissue codes (see **OME_TISSUE_CODE** and Section 1).

2.2 Identifiers and joins

- **pid** is the participant-level identifier and is used for participant-level joins across clinical/phenotype datasets and summaries.
- **vialLabel** is the biospecimen/sample identifier used as:
 - column names in Level 2 qc-norm matrices (***_QC\$qc_norm**),
 - the join key between sample-level metadata and the clinical/phenotype table **pheno\$data**.
- **feature_id** is the feature identifier used for feature-level joins:
 - qc-norm matrices use an assay-dependent feature identifier as rownames (see Section 5.2.1),
 - DA tables include a **feature_id** column,
 - the reference mapping **HUMAN_FEATURE_TO_GENE** is keyed by **feature_id** (and **assay**).

Note on casing: the consortium release notes prefer snake_case (e.g., **vial_label**), but existing packaged objects may also use camelCase variants (e.g., **vialLabel**) depending on file provenance.

2.3 Sample sizes

As summarized in Section 1 (Scope & scale):

- 175 unique participants (`pid`) are represented.
- 13,778 unique samples (`vialLabel`)
- Participant group counts (unique `pid`, adult sedentary cohort, baseline visit): `ADUResist` = 73, `ADUEndur` = 65, `ADUControl` = 37.
- Sex distribution for these 175 participants (from `pheno$data$sex_pzca`): Female = 126, Male = 49.
- Ages for these 175 participants (from `pheno$data$calculatedAge`): min 18, median 41, max 73.

3. Stats (Tissues, Assays, Platforms)

This section summarizes which types of analysis products are expected per omics layer for human-precovid-sed-adu v1.2 based on the consortium release readme and packaged objects.

Omics layer	Level 2 <code>qc-norm</code>	Level 3 <code>da</code>	Level 4 enrichment / integrative	Notes
Epigenomics (ATAC-seq, methyl-cap)	Yes (Cluod storage only)	Yes	Yes	Level 2 epigenomics is documented as analysis-hub files on GCS (see consortium readme); there are no packaged qc-norm epigenomics objects in data packages
Transcriptomics (RNA-seq)	Yes	Yes	Yes	R packaged objects include <code>*_TRNSCRPT_QC</code> (Level 2) and <code>*_TRNSCRPT_DA</code> (Level 3); integrative/enrichment summaries include transcriptomics in <code>CAMERA_RESULTS</code> and <code>FCM_ORA</code> .
Proteomics (Olink, global, phospho)	Yes	Yes	Yes	Packaged objects include <code>*_PROT_{OL, PR, PH}_QC</code> and corresponding <code>*_DA</code> tables; enrichment results include proteomics in <code>CAMERA_RESULTS</code> / <code>FCM_ORA</code> (assay-dependent).
Metabolomics (targeted <code>[T]</code> + untargeted <code>[U]</code>)	Yes	Yes	Yes	Multiple metabolomics platforms are provided as <code>*_METAB_T_*_QC</code> and <code>*_METAB_U_*_QC</code> plus consolidated <code>*_METAB_DA</code> ; metabolomics CV resources

Omics layer	Level 2 qc-norm	Level 3 da	Level 4 enrichment / integrative	Notes
				support redundancy handling (see Section 4 known issues).
Integrative / enrichment (cross-omics summaries)	N/A	N/A	Yes	CAMERA_RESULTS, FCM_ORA, FCM_CAMERA, FCM_CLUSTERS, and MOLECULAR_SIGNATURES provide Level 4 interpretation resources.

4. Version History & Known Issues

Versioning matters: This collection is released as a sequence of internal QC data freezes (**v1.0**, **v1.1**, **v1.2**, **v1.3**). For reproducible analyses and consistent sample/feature handling, reference the QC freeze version (e.g., **v1.3**) and its known issues/limitations. Note that R package versions are not the same as QC freeze versions.

Version	Date	Affected products	Key changes	Known issues / limitations	Recommendation
v1.3	2025-05-01	QC-normalized data, QC reports/metadata	Repaired metabolomic outlier and redundancy issues. See publication methods for details	N per group varies widely, recommend checking summary stats for details	Prefer v1.2 for current internal analysis; follow the guidance below for metabolomics and naming caveats.
v1.2	2024-12-20	QC-normalized data, QC reports/metadata, clinical data (internal release scope)	Data completed, such that all tissues/omes represented	See known issues below; in particular, metabolomics outlier handling and removed-samples table handling are in flux in v1.2.	Use only when reproducing prior internal analyses tied to v1.2.
v1.1	2024-04-01	QC-normalized data and associated artifacts		Data incomplete, released for early access	Use only when reproducing prior internal analyses tied to v1.1.

Version	Date	Affected products	Key changes	Known issues / limitations	Recommendation
v1.0	2024-02-05	QC-normalized data and associated artifacts		Data incomplete, released for early access	Use only when reproducing prior internal analyses tied to v1.0.

4.1 Known issues, limitations, and caveats (v1.2)

The consortium release readme describes the following issues that are relevant for analysis use (Levels 2–4):

- **Identifier/name casing inconsistencies:** key identifiers may appear with mixed casing/styles across files (e.g., `vialLabel`, `vial_label`, `labelID`, `label_id`); prefer `snake_case` where available but expect legacy variants.
- **Feature-level missingness / low N:** feature-level missingness varies; caution is advised for adipose proteomics and phosphoproteomics where some timepoint × group bins can have low effective N for DA modeling.

5. Data Access and Use

5.1 Data Access

- **Data Portal:** <https://motrpac-data.org> (after login, click “Pre-COVID Human Data Visualization”).
- **R packages (recommended for methods + convenient loading):**
 - `MotrpacHumanPreSuspensionAnalysis` (release pending, DA objects (Level 3) and Level 4 resources).
- **Source code / pipelines:**
 - `MotrpacHumanPreSuspension_packages/precovid-analyses/` (internal analysis code repository mirror).
 - `MotrpacHumanPreSuspension_packages/motrpac-bic-norm-qc/` (QC/normalization methods and reports repository mirror).
- **Supplementary methods:** see the publication for methods

5.2 Data Use

5.2.1 Object naming patterns (R packages and GCS)

This collection is distributed via both versioned files on the Data Hub and curated R data objects.

File naming via Data Hub (analysis-level products)

PreCAWG documentation describes analysis filenames that include the study code, a tissue-code token, an assay token, a product token (qc-norm/da/metadata), and a version suffix, e.g.:

```
human-precovid-sed-adu_[tissue-code]_[assay]_[qc-norm|da|metadata_*]_v<freeze>.txt
```

R data-object naming (packaged objects in **MoTrpacHumanPreSuspensionAnalysis**)

Most packaged analysis objects use a compact, uppercase naming convention:

[TISSUE]_[ASSAY]_[QC|DA]

Where:

- **[TISSUE]** is one of: **ADIPOSE**, **BLOOD**, **MUSCLE** (these are broad categories that map from MoTrPAC tissue codes; see **OME_TISSUE_CODE**).
- **[ASSAY]** is typically one of:
 - **TRNSCRPT** (RNA-seq),
 - **PROT_OL**, **PROT_PR**, **PROT_PH** (proteomics),
 - **METAB_T_*** and **METAB_U_*** (platform-specific metabolomics qc-norm),
 - **METAB** (consolidated metabolomics DA tables).
- **[QC|DA]** indicates Level 2 qc-norm (***_QC**) versus Level 3 DA (***_DA**).

5.2.2 Key objects / products (selected)

Selected high-value analysis objects (use **data(OBJECT_NAME)** to load when installed; see help pages for object-specific details):

Omics type / purpose	Object (pattern or example)	What it is	Dataset structure (high level)	Keys / join strategy
Participant/sample metadata	pheno	Curated phenotype/clinical data packaged as a motrdat object (data + dict)	pheno\$data is a long table keyed by vialLabel and pid with visit/timepoint and clinical fields	Join sample-level omics by vialLabel ; join participant-level summaries by pid
Feature mapping	HUMAN_FEATURE_TO_GENE	Cross-omics feature map for human features tested in DA	Table keyed by (assay , feature_id) with gene/protein/metabolite mappings	Join DA results by feature_id (and assay when needed)
Level 2 qc-norm (transcriptomics)	MUSCLE_TRNSCRPT_QC	RNA-seq qc-norm matrix + metadata (muscle category)	qc_norm matrix + sample_metadata + feature_metadata	Sample join by vialLabel ; feature join by feature_id
Level 3 DA (transcriptomics)	MUSCLE_TRNSCRPT_DA	RNA-seq DA table (muscle category)	Long-format table with effect sizes, CIs, p/FDR for	Feature join by feature_id ; annotate by

Omics type / purpose	Object (pattern or example)	What it is	Dataset structure (high level)	Keys / join strategy
			predefined contrasts	<code>contrast_short</code> , <code>Timepoint</code> , <code>randomGroupCode</code>
Level 2 qc-norm (proteomics)	<code>BLOOD_PROT_OL_QC</code>	Olink proteomics qc-norm matrix + metadata (blood category)	<code>qc_norm</code> matrix + metadata	Sample join by <code>vialLabel</code> ; feature join by <code>feature_id</code> (assay-dependent)
Level 3 DA (proteomics)	<code>BLOOD_PROT_OL_DA</code>	Olink proteomics DA table (blood category)	Long-format DA table	Feature join by <code>feature_id</code>
Level 2 qc-norm (metabolomics, platform-specific)	<code>BLOOD_METAB_URPNEG_QC</code> (and other <code>*_METAB_{T,U}*_*_QC</code>)	Platform-specific metabolomics qc-norm matrix + metadata	<code>qc_norm</code> matrix + metadata	Sample join by <code>vialLabel</code> ; metabolomics feature handling may use <code>METABOLOMICS_CVS / HUMAN_FEATURE_TO_GENE</code>
Level 3 DA (metabolomics, consolidated)	<code>BLOOD_METAB_DA</code>	Consolidated metabolomics DA table (blood category)	Long-format DA table across metabolomics	Feature join by <code>feature_id</code> (RefMet-aware handling recommended; see <code>motrpac-bic-norm-qc</code> and <code>precovid-analyses</code> for implementation examples)
Level 4 enrichment (CAMERA)	<code>CAMERA_RESULTS</code>	Enrichment results organized by tissue/assay/contrast	Long-format enrichment results table	Join to DA context by <code>tissue</code> , <code>assay</code> , <code>contrast_short</code>
Level 4 enrichment (clustering ORA)	<code>FCM_ORA</code> (plus <code>FCM_CAMERA</code> , <code>FCM_CLUSTERS</code>)	Cluster-based enrichment and clustering results	Tables (ORA) plus list objects (clusters)	Join strategy varies; typically by

Omics type / purpose	Object (pattern or example)	What it is	Dataset structure (high level)	Keys / join strategy
				tissue, assay, cluster identifiers

5.2.3 Data Object Structure and Definitions

5.2.3.1 QC Norm objects

Common qc-norm object structure for *_QC objects:

- *_QC\$qc_norm: a `data.frame` with features as rownames and `vialLabel` values as column names (row identifiers are assay-dependent).
- *_QC\$sample_metadata: a `data.frame` keyed by `vialLabel` (may also include legacy variants such as `vial_label` and `label_id` depending on provenance).
- *_QC\$feature_metadata: a `data.frame` with assay-dependent feature descriptors (e.g., transcript `feature_id`, proteomics target IDs, metabolite descriptors such as `metabolite_name/mz/rt`).

5.2.3.2 DA objects

Below is the data dictionary for a standard DA object

column name	Alias (Field Name description)	Data Type	Categorical Values	Categorical Definitions
tissue	name of the tissue (muscle, adipose, blood) being tested	character	muscle adipose blood	If the tissue being investigated came from a muscle, adipose, or blood tissue. These tissue categories encompass tissue subclasses - e.g. PBMCs will be categorized under blood
assay	name of the assay (ie prot_pr, etc) being tested	character	metab transcript-rna-seq prot-ol prot-ph prot-pr	The assay that is undergoing hypothesis testing. All metabolomics platforms are listed under metab. See the platform column for more details
platform	Specific metabolomic platform that is being tested	factor	metab-u-hilicpos metab-u-ionpneg metab-u-lrpneg	For entries that are under the metab assay category, this

			metab-u-lrppos metab-u-rpneg metab-u-rppos metab-t-acoa metab-t-amines metab-t-ka metab-t-nuc metab-t-oxyipneg metab-t-tca metab-t-conv <NA>	column describes specifically which metabolomic platform the entry is from. NA for all non-metabolite platforms.
full_model	A formal specification of the model, ie: ~ 0 + group_timepoint + sex_psca + age + BMI + (1 PID)	character		Includes information about how the linear model was implemented
contrast	A formal specification of the contrast, ie: (ADUResist_post_15_min - ADUResist_pre_exercise) -(ADUControl_post_15_min - ADUControl_pre_exercise)	factor	See methods- 33 possible comparisons	Includes comparison of the fixed effects from the 'full_model' entry that are formally being tested
contrast_short	A shorter version of the contrast column	factor	see methods - still 33 possible comparisons	Connects 1:1 with entries from contrast. Included for ease of use. Potentially better for figure panels.
contrast_type	Annotates which group comparisons are being made.	factor	exercise_with_controls exercise_no_controls Endur_vs_Resist baseline control_only	Helpful for filtering for comparisons of interest
contrast_category	Annotates which specific exercise groups are being compared	factor	EE-CON RE-CON EE-EE RE-RE EE-RE CON-CON	EE indicates the Endur group, RE indicates Resistance group, CON indicates the Control group
randomGroupCode	For the first element of the contrast: the random group code, which indicates modality	character	ADUControl ADUEndur ADUResist	EE indicates the Endur group, RE indicates Resistance group, CON indicates the Control group. Direct EE, RE comparisons are categorized under ADUEndur in this case
Timepoint	For the first element of the contrast: Identifier of the biospecimen collection timepoint	varchar	pre_exercise during_20_min during_40_min post_10_min post_15_min	Timepoint of comparison. For comparisons relative to pre-exercise, the exercise/post

			post_30_min post_45_min post_4_hr post_24_hr	exercise timepoint is highlighted.
feature_id	a unique identifier of the feature. Will be ome specific	character		Name of the feature being tested. For example, in the transcript-rna-seq assay, this will correspond to an ensemble gene name. See the HUMAN_FEATURE_TO_GENE file in the R package or motrpac data hub for more information about how some gene-level features are mapped to genes.
logFC	log Fold Change. Represents the effect size.	Numeric		
CI.L	Left bound of the logFC Confidence interval	Numeric		
CI.R	Right bound of the logFC Confidence interval	Numeric		
degrees_of_freedom	Residual Degrees of freedom	Numeric		
logLik	Log Likelihood	Numeric		
t	T-statistic; direct output of the dream model	Numeric		
z.std	Standardized Z-Score; direct output of the dream model.	Numeric		
AveExpr	Mean expression of the input. In the units of the relevant assay category.	Numeric		
p_value	p-value; direct output of the dream model	Numeric		
adj_p_value	Benjamini Hotchberg adjustment of the p-value column	Numeric		

5.2.4 Example R workflow (qc-norm + DA + pheno join)

```
None
library(MotrpacHumanPreSuspensionAnalysis)
```

```

# Sample metadata / phenotype (packaged as a motrdat object)
data(pheno)

# Example Level 2 qc-norm object and Level 3 DA table
data(MUSCLE_TRNSCRPT_QC)
data(MUSCLE_TRNSCRPT_DA)

# Optional: feature mapping (useful for annotation)
data(HUMAN_FEATURE_TO_GENE)

# 1) Extract a qc-norm matrix and its sample IDs (vialLabel)
expr <- MUSCLE_TRNSCRPT_QC$qc_norm
sample_ids <- colnames(expr)

# 2) Join sample IDs to phenotype table (pheno$data is keyed by vialLabel)
ph <- pheno$data
ph_sub <- ph[match(sample_ids, ph$vialLabel), c("pid", "randomGroupCode", "Timepoint",
"visitcode", "sex_pzca", "calculatedAge")]
stopifnot(!any(is.na(ph_sub$pid)))

# 3) Inspect a DA contrast and annotate a few top hits
contrast_name <- "Endur.post_24_hr - Control.post_24_hr (delta-delta)"
da_sub <- subset(MUSCLE_TRNSCRPT_DA, contrast_short == contrast_name)
da_sub <- da_sub[order(da_sub$adj_p_value), ]

# Optional: add gene symbols (assay-dependent; for transcriptomics this is typical)
annot <- HUMAN_FEATURE_TO_GENE[HUMAN_FEATURE_TO_GENE$assay == "transcript-rna-seq",
c("feature_id", "gene_symbol")]
da_annot <- merge(da_sub, annot, by = "feature_id", all.x = TRUE)

head(da_annot[, c("feature_id", "gene_symbol", "logFC", "p_value", "adj_p_value")])

```

6. Citations, Licensing, and Contact

- **Citation / primary documentation:** MoTrPAC Bioinformatics Center (BIC). *Consortium Release: human-precovid-sed-adu-v1.3*
- **Licensing / restrictions:** Open data are distributed via CC BY 4.0 license. Individual data controlled by data access restrictions are covered by relevant Data Access restrictions. Package source code for `MotrpacHumanPreSuspension` and `MotrpacHumanPreSuspensionData` is distributed under MIT (see each package's [LICENSE](#)).
- **Contact/support:** motrpac-helpdesk@lists.stanford.edu

Data Processing and Analysis Framework

The BIC ingests and disseminates a wide range of data for users. Below is an outline of the general approach taken to molecular data. Similar structure exists for clinical/phenotypic data, though there are fewer levels of processing (See Clinical/Phenotypic data collections)

Levels 0-1: Raw and Processed Data

Raw data (Level 0) consists of unprocessed instrumental output that cannot be directly analyzed. Processing converts this raw data into quantifiable measurements (Level 1). Depending on the assay, this processing may occur at the data generation site or at the BIC

Level 1 datasets represent the first layer of biologically interpretable information. These are direct outputs from standardized bioinformatics pipelines specific to each omics platform:

- **Transcriptomics:** BAM files containing aligned sequencing reads are converted to gene expression matrices, where values represent read counts for each gene in each sample
- **Proteomics:** Raw mass spectrometry files are processed to identify peptides and proteins from spectral signatures and quantify their abundance
- **Metabolomics:** Instrumental data are processed to identify and quantify metabolite levels

 Typical data object formats:

- A structured feature-by-sample matrix where rows represent biological entities (genes, proteins, metabolites) and columns represent individual samples. These datasets serve as quality-controlled inputs for downstream analysis.
- Metadata tables for features and samples. These vary by assay

Level 2: Quality-Controlled Normalized Data

Level 2 data are normalized, batch-corrected, and filtered to remove outliers. **This level is analysis-specific** because the additional steps require consideration of the experimental design and scientific question.

Platform-specific normalization methods are applied:

- **Transcriptomics:** TMM normalization to account for sequencing depth differences
- **Proteomics and Metabolomics:** Median absolute deviation (MAD) scaling
- **Metabolomics:** Missing value imputation is commonly applied
- **Proteomics:** Missing values are typically not imputed

 Typical data object formats:

- A structured feature-by-sample matrix where rows represent biological entities (genes, proteins, metabolites) and columns represent individual samples. These datasets serve as analysis specific, individual level data inputs for downstream analysis.
- Metadata tables for features and samples. These vary by assay.

Level 3: Differential Abundance Analysis

Differential analysis identifies variations in molecular abundance between experimental groups. This provides analyte-specific estimates of experimental effects. Statistical methods are selected based on experimental design and may include:

- Contrasts to control measurements
- Mixed models for repeated measures
- Multiple comparison correction
- Covariate adjustment



Typical data object formats:

- A differential abundance (DA) table typically has one feature analyzed using one model per row. Columns in this table fall into three general categories
 - Feature metadata (e.g. omic layer, tissue source)
 - Model metadata (e.g. mixed effects model, covariats, contrast of interest)
 - Summary statistics (e.g. beta coefficient, confidence intervals, p-value, adjusted p-value)

Level 4: Multi-Omics Integration and Biological Interpretation

Multi-omics integration methods vary based on research objectives and typically employ data science approaches including feature reduction, clustering, and network analysis.

Multi-omics analysis leverages existing biological knowledge for enhanced interpretation:

- **Pathway enrichment analysis** identifies biological processes modulated by the intervention using Gene Set Enrichment Analysis (GSEA), over-representation analysis, and databases including Gene Ontology, KEGG, and Reactome
- **Transcription factor analysis** uses genomic motif analysis tools (e.g., HOMER) to identify regulatory factors controlling observed expression changes
- **Gene regulatory network modeling** applies specialized tools to model regulatory relationships from multi-omics data

Cross-species comparative analysis enables translation between animal models and human studies by:

- Mapping orthologous genes and proteins between species
- Identifying conserved molecular signatures and pathways
- Generating translational hypotheses for human health applications



Typical data object formats:

- Pathway enrichment results:
 - Pathway metadata, enrichment statistics
- Transcription factor analysis
 - Transcription factor gene symbol, enrichment statistics
- Gene regulatory network
 - Edge and node identifications and metrics

